Single Random Variable Distributions

A complete reference: intuition, formulas, and problem-solving strategies

1. Foundational principle: story first, formula second

Every distribution has a generative narrative—a physical or combinatorial process that produces it. If you can describe that process in plain language, the formula becomes a consequence rather than a thing to memorise. The PMF/PDF is simply a precise accounting of the story. When approaching an unfamiliar distribution, ask first: *what kind of experiment or process would naturally produce this shape?*

> **Reverse-engineering direction:** describe the process → derive the distribution. This is always more durable than memorising formulas in the forward direction.

2. The discrete family

The Bernoulli backbone

Everything flows from a single binary trial. Fix n trials and count successes: Binomial. Ask how many trials until the first success: Geometric (the only discrete memoryless distribution). Ask how many trials until the r-th success: Negative Binomial. This entire family shares one underlying probability space—the infinite sequence of i.i.d. Bernoulli trials—and each distribution is a different question asked of that same sequence.

A critical relationship: $\text{Binomial}(n, p) \to \text{Poisson}(np)$ as $n \to \infty$ with $np = \lambda$ fixed. The Binomial→Normal CLT limit runs in the other direction: large n with fixed p produces $\mathcal{N}(np,\, np(1-p))$.

Poisson: the events-in-time engine

Stands independently—it characterises events at a constant rate in continuous time or space. Its defining signature is Mean = Var = λ. Whenever observed variance $\gg$ mean in count data, the Poisson is inadequate—use Negative Binomial (which has a Poisson–Gamma mixture interpretation).

Hypergeometric: the without-replacement wrinkle

When sampling without replacement from a finite population, the independence assumption of Binomial breaks down. The finite population correction $(N-n)/(N-1)$ reduces variance. When $n \ll N$, the correction $\to 1$ and Hypergeometric $\approx \text{Binomial}(n, K/N)$.

3. The continuous family

The Exponential–Gamma spine

Exponential(λ) is the waiting time for the first Poisson event and the unique continuous memoryless distribution. Summing n independent copies gives $\text{Gamma}(n, \lambda)$. The special case $\text{Gamma}(k/2, 1/2) = \chi^2(k)$ connects this spine directly to the Normal world.

The Normal and its court

The CLT makes Normal the universal attractor. Squaring and summing k standard Normals gives $\chi^2(k)$. Forming $Z/\sqrt{V/\nu}$ with $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2(\nu)$ gives Student's t. A ratio of two chi-squareds-over-df gives the F distribution. The entire apparatus of classical hypothesis testing lives in this derived family.

Beta: distributions over probabilities

The only family naturally on $[0, 1]$. The Bayesian update rule $\text{Beta}(\alpha, \beta) \to \text{Beta}(\alpha + k,\ \beta + n - k)$ is the cleanest expression of learning from data. The sum $\alpha + \beta$ is the effective sample size of the prior.

Heavy-tailed alternatives

Log-Normal arises from multiplicative processes. Weibull generalises Exponential with shape k: $k < 1$ (decreasing hazard), $k = 1$ (Exponential), $k > 1$ (increasing hazard). Cauchy is the pathological case—symmetric and bell-shaped but with no moments, and the CLT fails completely.

4. Cross-cutting strategies

1. **Build and internalize the relationships map.** The distributions form a network, not a list. Bernoulli $\to$ Binomial $\to$ Poisson $\to \mathcal{N}$; Exp $\to$ Gamma $\to \chi^2 \to t, F$; Unif $\to$ Beta via order statistics. Drawing and redrawing this network is the highest-leverage single activity.

2. **Diagnose distributions from problem text.** "Number of successes in n trials" $\to$ Binomial. "Waiting time until first event" $\to$ Exponential or Geometric. "Count of events in a fixed window" $\to$ Poisson. "Proportion or probability on $[0, 1]$" $\to$ Beta. "Time to failure with wear-out" $\to$ Weibull. "Comparing two variances" $\to$ F distribution.

3. **Derive mean and variance from first principles at least once.** Derive $E[X] = np$ for Binomial from $E[X] = \sum_k k\, P(X = k)$ using the indicator variable trick $X = X_1 + \cdots + X_n$, $X_i \sim \text{Bernoulli}(p)$. This makes formulas unforgettable and teaches a technique that transfers to every other distribution.

4. **Master the MGF as the unifying tool.** $M(t) = E[e^{tX}]$. Two facts: (a) if X and Y are independent, $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$—this proves sums of Normals are Normal, sums of Poissons are Poisson; (b) the k-th derivative at $t = 0$ gives $E[X^k]$. Deriving the Normal MGF $M(t) = \exp\!\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$ once makes every sum-of-Normals problem trivial.

5. **Study parameter sensitivity experimentally in R.** For every distribution, vary each parameter and plot the density. For Gamma: vary α from 0.5 to 10 and watch the L-shape transition to a bell. For Beta: vary α and β across $\{0.5,\ 1,\ 2,\ 10\}$ and observe U-shapes, uniform, skewed, and symmetric modes. 20 minutes of plotting internalises more than hours of formula study.

6. **Exploit limiting relationships as computational tools.** $\text{Binomial}(1000,\ 0.001) \approx \text{Poisson}(1)$. $\text{Poisson}(100) \approx \mathcal{N}(100, 100)$. $t(60) \approx \mathcal{N}(0, 1)$. $\text{Gamma}(50, 1) \approx \mathcal{N}(50, 50)$.

7. **Know the tail behaviour by rote.** Normal tails: $\exp(-x^2/2)$ (very fast). Exponential: e^{-x} (fast). Log-Normal: between exponential and power-law. Cauchy/Pareto: polynomial decay (catastrophically slow). This hierarchy determines behaviour under extremes.

8. **Use characterisation theorems.** Memorylessness $\leftrightarrow$ Exponential (continuous) or Geometric (discrete). Maximum entropy on a bounded interval $\leftrightarrow$ Uniform. Maximum entropy subject to fixed mean and variance $\leftrightarrow$ Normal. Checking which properties your physical system satisfies often identifies the distribution uniquely.

5. Problem recognition heuristics

Problem cue	**Distribution**
Binary outcomes, fixed n trials	Binomial(n, p)
How long until first (or r-th) success	Geometric or Negative Binomial
Count of events in continuous interval, constant rate	Poisson; inter-arrivals $\sim$ Exp(λ)
Sampling without replacement from finite population	Hypergeometric
Proportion, probability, or rate on $[0, 1]$	Beta
Ratio of two sample variances, ANOVA	F distribution
Sample mean from Normal population, unknown σ	Student's t
Positive quantity, multiplicatively generated	Log-Normal or Gamma
Time to failure, wear-out or infant mortality	Weibull
Symmetric, no moments, CLT fails	Cauchy

6. Distribution reference cards — Discrete family

Bernoulli

Discrete

$X \sim \text{Bernoulli}(p)$

A single experiment with two outcomes: success (1) with probability p, failure (0) with probability $1-p$. The atomic unit of all binary randomness.

$$P(X=k) \;=\; p^k(1-p)^{1-k}, \qquad k \in \{0,1\}$$

Mean p

Variance $p(1-p)$

Parameters:

- $p \in (0,1)$: probability of success

> **Key insight:** Variance is maximised at $p = \frac{1}{2}$ and vanishes as $p \to 0$ or $p \to 1$. The indicator variable technique—writing any random variable as a sum of Bernoullis—is one of the most powerful shortcuts in discrete probability.

R:

```
rbinom(n=100, size=1, prob=0.3)  # size=1 gives Bernoulli
p <- seq(0,1,.01); plot(p, p*(1-p), type='l')
```

Connections: $\to$ Binomial: sum of n i.i.d. copies • $\to$ Geometric: trials until first success • $\to$ Negative Binomial: trials until r successes

Binomial

Discrete

$X \sim \text{Binomial}(n, p)$

Count of successes in n independent Bernoulli(p) trials. The fundamental distribution of counts with a fixed number of independent binary chances.

$$P(X=k) \;=\; \binom{n}{k} p^k(1-p)^{n-k}, \qquad k = 0,1,\ldots,n$$

Mean np

Variance $np(1-p)$

Parameters:

- $n \in \mathbb{N}$: number of trials
- $p \in (0,1)$: success probability per trial

Key insight: $\binom{n}{k}$ counts arrangements; $p^k(1-p)^{n-k}$ is the probability of any specific one. Two limits: as $n \to \infty$ with $np = \lambda$ fixed, Binomial $\to$ Poisson(λ). By the CLT, as $n \to \infty$ with fixed p, Binomial $\to \mathcal{N}(np, np(1-p))$. These two limits tell you when to approximate.

R:

```
rbinom(1000, size=20, prob=0.3)
plot(table(rbinom(5000, 20, 0.3)))  # visualise pmf
```

Connections: Sum of n i.i.d. Bernoulli(p) • $\to$ Poisson(λ) as $n \to \infty$, $np = \lambda$ • $\to \mathcal{N}(np, np(1-p))$ by the CLT

Geometric

Discrete

$X \sim \text{Geometric}(p)$

Number of Bernoulli(p) trials needed to get the first success. Two conventions exist: include the successful trial (support $\{1, 2, \ldots\}$) or count failures before success (support $\{0, 1, \ldots\}$).

$$P(X = k) \;=\; (1-p)^{k-1}\,p, \qquad k = 1, 2, 3, \ldots$$

Mean $\dfrac{1}{p}$

Variance $\dfrac{1-p}{p^2}$

Parameters:

- $p \in (0, 1)$: success probability per trial

Key insight: The only discrete memoryless distribution: $P(X > m + n \mid X > m) = P(X > n)$. Having already waited m trials gives no information about future waiting time. This property characterises it uniquely—just as the Exponential is the unique memoryless continuous distribution.

R:

```
rgeom(1000, prob=0.2)  # R: failures before success convention
mean(rgeom(100000, 0.2))  # verify (1-p)/p = 4
```

Connections: Special case of Neg. Binomial ($r = 1$) • Discrete analog of Exponential • $\to$ Exponential in the continuous-time limit

Negative Binomial

Discrete

$X \sim \text{NegBin}(r, p)$

Number of failures before the r-th success in a sequence of Bernoulli(p) trials. Generalises Geometric ($r = 1$) and is the natural model for overdispersed count data.

$$P(X = k) \;=\; \binom{k+r-1}{k} p^r (1-p)^k, \qquad k = 0, 1, 2, \ldots$$

Mean $\dfrac{r(1-p)}{p}$

Variance $\dfrac{r(1-p)}{p^2}$

Parameters:

- $r > 0$: required number of successes
- $p \in (0, 1)$: success probability per trial

Key insight: Key identity: a Poisson(λ) where $\lambda \sim$ Gamma(r, $p/(1-p)$) produces Negative Binomial. This Poisson–Gamma mixture is the signature of overdispersion: Var / Mean $= 1/p > 1$, unlike Poisson where Var = Mean.

R:

```
rnbinom(1000, size=3, prob=0.4)  # size = r
x <- rnbinom(10000, 3, .4); var(x)/mean(x)  # > 1 (overdispersed)
```

Connections: $r = 1$: Geometric(p) • Poisson–Gamma mixture (overdispersion) • $\to$ Poisson as $p \to 1$ with $r(1-p)/p = \lambda$ fixed

Poisson

Discrete

$X \sim$ Poisson(λ)

Count of events in a fixed interval when events occur at constant rate λ, independently, with no clustering. The canonical model for rare events: radioactive decays, mutations, call arrivals.

$$P(X = k) \;=\; \frac{e^{-\lambda}\,\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Mean λ

Variance λ

Parameters:

- $\lambda > 0$: mean event rate per unit time or space

Key insight: Mean = Var = λ is the key diagnostic: observed Var $\gg$ Mean signals overdispersion—use Negative Binomial instead. The Poisson process generates Exponentially distributed inter-arrival gaps. As $\lambda \to \infty$, Poisson(λ) $\to \mathcal{N}(\lambda, \lambda)$ by the CLT.

R:

```
rpois(1000, lambda=5)
x <- rpois(10000, 5); c(mean(x), var(x))  # both approx lambda
```

Connections: Limit of Binomial(n, p) as $n \to \infty$, $np = \lambda$ • Inter-arrivals $\sim$ Exp(λ) • $\to \mathcal{N}(\lambda, \lambda)$ as $\lambda \to \infty$

Hypergeometric

Discrete

$X \sim \text{HyperGeom}(N, K, n)$

Drawing n items without replacement from a population of N containing K successes. Models quality-control sampling, ecological capture–recapture, and Fisher's exact test.

$$P(X = k) \;=\; \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

Mean $\dfrac{nK}{N}$

Variance $n \cdot \dfrac{K}{N} \cdot \left(1 - \dfrac{K}{N}\right) \cdot \dfrac{N-n}{N-1}$

Parameters:

- $N \in \mathbb{N}$: population size
- $K \leq N$: successes in population
- $n \leq N$: number drawn

Key insight: The factor $(N-n)/(N-1)$ is the finite population correction: it reduces variance when sampling a large fraction of the population. When $n \ll N$, the correction $\to 1$ and HyperGeom $\approx$ Binomial$(n, K/N)$—without replacement approximates with replacement for small samples.

R:

```
rhyper(nn=1000, m=K, n=N-K, k=n)  # m=K successes in pop
```

Connections: $\to$ Binomial$(n, K/N)$ as $N \to \infty$ • Basis of Fisher's exact test • Without-replacement sampling structure

Discrete Uniform

Discrete

$X \sim \text{DUnif}(a, b)$

Each integer in $\{a, a+1, \ldots, b\}$ is equally likely. The die-roll distribution and the maximum-entropy distribution over a finite integer set.

$$P(X = k) \;=\; \frac{1}{b-a+1}, \qquad k \in \{a, a+1, \ldots, b\}$$

Mean $\dfrac{a+b}{2}$

Variance $\dfrac{(b-a+1)^2-1}{12}$

Parameters:

- $a, b \in \mathbb{Z},\ a < b$: lower and upper integer bounds

> **Key insight:** Variance $= (n^2-1)/12$ where $n = b-a+1$, directly analogous to Continuous Uniform's $(b-a)^2/12$. Maximises entropy over $\{a, \ldots, b\}$. All discrete distributions are simulable from Discrete Uniform via look-up tables—this is the basis of the inverse CDF simulation method.

R:

```
sample(a:b, size=1000, replace=TRUE)
x <- sample(1:6, 100000, TRUE); c(mean(x), var(x))  # 3.5, 2.92
```

Connections: Maximum entropy on bounded integers

- Analogous to Continuous Uniform
- Engine of all discrete simulation

7. Distribution reference cards — Continuous family

Uniform

Continuous

$X \sim \text{Uniform}(a, b)$

Every point in $[a, b]$ is equally likely. Maximum-entropy distribution on a bounded interval and the universal source for simulation via the probability integral transform.

$$f(x) = \frac{1}{b-a}, \quad x \in [a, b]; \qquad F(x) = \frac{x-a}{b-a}$$

Mean $\dfrac{a+b}{2}$

Variance $\dfrac{(b-a)^2}{12}$

Parameters:

- $a < b$: lower and upper bounds

Key insight: Probability integral transform: if F is any CDF and $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U)$ has CDF F. This is why all simulation starts with `runif()`. The minimum and maximum of n i.i.d. Uniform$(0, 1)$ variables follow Beta distributions; $\text{Unif}(0, 1) = \text{Beta}(1, 1)$.

R:

```
runif(1000, min=a, max=b)
qnorm(runif(10000))  # generates N(0,1) via inverse CDF
```

Connections: $\text{Unif}(0, 1) = \text{Beta}(1, 1)$ • Maximum entropy on bounded interval • Engine of all Monte Carlo simulation

Exponential

Continuous

$X \sim \text{Exponential}(\lambda)$

Waiting time until the first event of a Poisson(λ) process. The only continuous memoryless distribution—the backbone of survival analysis and queueing theory.

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0; \qquad F(x) = 1 - e^{-\lambda x}$$

Mean $\dfrac{1}{\lambda}$

Variance $\dfrac{1}{\lambda^2}$

Parameters:

- $\lambda > 0$: rate — events per unit time

Key insight: Memorylessness: $P(X > s + t \mid X > s) = P(X > t)$—unique among continuous distributions. Minimum of n independent $\text{Exp}(\lambda_i)$ is $\text{Exp}(\sum \lambda_i)$. Sum of n i.i.d. $\text{Exp}(\lambda)$ is $\text{Gamma}(n, \lambda)$. Two parameterisations (rate vs. scale) exist—always verify which `R` uses.

R:

```
rexp(1000, rate=lambda)
x <- rexp(50000, 1); mean(x[x>2] - 2)  # verify memorylessness: approx 1
```

Connections: Inter-arrival times of Poisson(λ) process • $\to$ Gamma(n, λ) as n-fold sum • Continuous analog of Geometric

Gamma

Continuous

$X \sim \text{Gamma}(\alpha, \beta)$

Waiting time until the α-th Poisson event. For integer α, the sum of α independent Exponential(β) variables. Generalises to non-integer α via the Gamma function.

$$f(x) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}, \qquad x > 0$$

Mean $\dfrac{\alpha}{\beta}$

Variance $\dfrac{\alpha}{\beta^2}$

Parameters:

- $\alpha > 0$: shape — controls skewness and event count
- $\beta > 0$: rate $= 1/extscale$

Key insight: Small α gives a heavy right tail; large $\alpha \to \mathcal{N}(\alpha/\beta,\, \alpha/\beta^2)$ by the CLT. Key special cases: $\text{Gamma}(1, \beta) = \text{Exp}(\beta)$; $\text{Gamma}(k/2,\, 1/2) = \chi^2(k)$. The Gamma function satisfies $\Gamma(n) = (n-1)!$ for positive integers. Two parameterisations exist (rate vs. scale)—`R`'s `rgamma` supports both.

R:

```
rgamma(1000, shape=alpha, rate=beta)
par(mfrow=c(2,2))
for(a in c(.5,1,3,10)) hist(rgamma(2000,a,1), main=paste('alpha=',a))
```

Connections: $\alpha = 1$: Exponential(β) • $\alpha = k/2$, $\beta = 1/2$: $\chi^2(k)$ • Sum of α i.i.d. $\text{Exp}(\beta)$

Normal

Continuous

$X \sim \mathcal{N}(\mu,\, \sigma^2)$

The CLT attractor: sums of independent finite-variance random variables converge in distribution

to Normal regardless of the original shape. Symmetric and completely characterised by mean and variance.

$$f(x) \;=\; \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}$$

Mean μ

Variance σ^2

Parameters:

- $\mu \in \mathbb{R}$: mean (location)
- $\sigma^2 > 0$: variance (spread)

Key insight: Standardisation: $Z = (X-\mu)/\sigma \sim \mathcal{N}(0,1)$. The 68–95–99.7 rule: $P(|Z|<1) \approx 0.68$, $P(|Z|<2) \approx 0.95$, $P(|Z|<3) \approx 0.997$. MGF $M(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$: independent Normals sum to Normal with additive parameters. Maximum entropy distribution subject to fixed mean and variance.

R:

```
rnorm(1000, mean=mu, sd=sigma)
means <- replicate(2000, mean(rexp(30,1)))
hist(means, freq=FALSE)  # CLT: Normal(1, 1/30)
```

Connections: CLT limit of any finite-variance sum • $\to$ Log-Normal via e^X • Generates χ^2, t, F via combinations

Log-Normal

Continuous

$X \sim \mathrm{LogN}(\mu, \sigma^2)$

If $\log X \sim \mathcal{N}(\mu, \sigma^2)$ *then* X *is Log-Normal. Arises from multiplicative processes—many small random multiplications, just as Normal arises from additions.*

$$f(x) \;=\; \frac{1}{x\,\sigma\sqrt{2\pi}}\exp\left(-\frac{(\ln x-\mu)^2}{2\sigma^2}\right), \qquad x > 0$$

Mean $e^{\mu+\sigma^2/2}$

Variance $\left(e^{\sigma^2}-1\right)e^{2\mu+\sigma^2}$

Parameters:

- $\mu \in \mathbb{R}$: mean of $\log X$
- $\sigma^2 > 0$: variance of $\log X$

Key insight: Mean $= e^{\mu+\sigma^2/2}$ always exceeds median $= e^{\mu}$: always right-skewed. This is why the arithmetic mean exceeds the geometric mean for positive data. Natural for: stock prices (multiplicative returns), incomes, particle sizes, biological quantities that grow multiplicatively.

R:

```
rlnorm(1000, meanlog=mu, sdlog=sigma)
hist(log(rlnorm(10000,0,1)))  # should look N(0,1)
```

Connections: $\log X \sim \mathcal{N}(\mu, \sigma^2)$ • Multiplicative CLT limit • Models asset prices, incomes, sizes

Beta

Continuous

$X \sim \text{Beta}(\alpha, \beta)$

A distribution over $(0, 1)$*. Models probabilities, proportions, and rates. The conjugate prior for Bernoulli/Binomial likelihoods in Bayesian statistics.*

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \qquad x \in (0, 1)$$

Mean $\dfrac{\alpha}{\alpha+\beta}$

Variance $\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Parameters:

- $\alpha > 0$: shape — think prior successes $+1$
- $\beta > 0$: shape — think prior failures $+1$

Key insight: Shape intuition: $\alpha = \beta = 1 \to$ Uniform; $\alpha = \beta > 1 \to$ symmetric bell at $\frac{1}{2}$; $\alpha > 1 > \beta \to$ right-skewed; $\alpha < 1$ or $\beta < 1 \to$ U-shaped. Bayesian update: $\text{Beta}(\alpha, \beta) + k$ successes in n trials $\to \text{Beta}(\alpha + k, \beta + n - k)$. The sum $\alpha + \beta$ is the effective sample size of the prior.

R:

```
rbeta(1000, shape1=alpha, shape2=beta)
curve(dbeta(x,1+7,1+3), 0,1)  # posterior after 7 heads in 10 flips
```

Connections: $\text{Unif}(0, 1) = \text{Beta}(1, 1)$ • Conjugate prior for Binomial • α-th order statistic of $(\alpha + \beta - 1)$ Uniforms

Chi-Squared

Continuous

$X \sim \chi^2(k)$

Sum of squares of k *independent standard Normal variables. The fundamental distribution for*

inference about variance, and the source of t and F distributions.

$$f(x) = \frac{x^{k/2-1}\, e^{-x/2}}{2^{k/2}\,\Gamma(k/2)}, \qquad x > 0$$

Mean k

Variance $2k$

Parameters:

- $k \in \mathbb{N}$: degrees of freedom

Key insight: $\chi^2(k) = \text{Gamma}(k/2, 1/2)$. As $k \to \infty$, $(\chi^2(k) - k)/\sqrt{2k} \to \mathcal{N}(0,1)$. The sample variance satisfies $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$—the $n-1$ d.f. encode the constraint $\sum(X_i - \bar{X}) = 0$. Additive: the sum of independent chi-squareds is chi-squared with summed degrees of freedom.

R:

```
rchisq(1000, df=k)
k<-5; x<-replicate(2000, sum(rnorm(k)^2))
hist(x, freq=FALSE); curve(dchisq(x,k), add=TRUE, col='red')
```

Connections: Sum of k squared $\mathcal{N}(0,1)$ variables • Gamma$(k/2, 1/2)$ special case • Generates t and F distributions

Student's t

Continuous

$X \sim t(\nu)$

Ratio $Z/\sqrt{V/\nu}$ where $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2(\nu)$ are independent. Arises when estimating a Normal mean with unknown variance replaced by S from data.

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad x \in \mathbb{R}$$

Mean $0 \quad (\nu > 1)$

Variance $\dfrac{\nu}{\nu - 2} \quad (\nu > 2)$

Parameters:

- $\nu > 0$: degrees of freedom — heavier tails as ν decreases

Key insight: As $\nu \to \infty$, $t(\nu) \to \mathcal{N}(0,1)$. At $\nu = 1$ it is the Cauchy—no moments exist. The t-test uses this because replacing σ with S introduces $\chi^2(\nu)$ randomness in the denominator. Key identity: $T \sim t(\nu) \Rightarrow T^2 \sim F(1,\nu)$.

R:

```r
rt(1000, df=nu)
hist(rt(10000,3), xlim=c(-8,8), main='t(3) vs N(0,1)')
hist(rnorm(10000), xlim=c(-8,8))  # compare tails
```

Connections: $\mathcal{N}(0,1)/\sqrt{\chi^2(\nu)/\nu}$ • $\to \mathcal{N}(0,1)$ as $\nu \to \infty$ • $\nu = 1$: Cauchy distribution

F Distribution

Continuous

$X \sim F(d_1, d_2)$

Ratio of two independent chi-squared variables each divided by their degrees of freedom. Used to compare two variances and in ANOVA and regression F-tests.

$$X = \frac{\chi^2(d_1)/d_1}{\chi^2(d_2)/d_2}, \qquad X > 0$$

Mean $\dfrac{d_2}{d_2 - 2} \quad (d_2 > 2)$

Variance $\dfrac{2d_2^2(d_1 + d_2 - 2)}{d_1(d_2 - 2)^2(d_2 - 4)} \quad (d_2 > 4)$

Parameters:

- $d_1 > 0$: numerator degrees of freedom
- $d_2 > 0$: denominator degrees of freedom

Key insight: Key identity: $T \sim t(\nu) \Rightarrow T^2 \sim F(1, \nu)$. Always positive and right-skewed. Under H_0: $\sigma_1^2 = \sigma_2^2$, the ratio $S_1^2/S_2^2 \sim F(n_1 - 1, n_2 - 1)$. Asymmetry: $F_\alpha(d_1, d_2) = 1/F_{1-\alpha}(d_2, d_1)$, so lower-tail critical values come for free.

R:

```r
rf(1000, df1=d1, df2=d2)
t_sq <- rt(10000,10)^2  # verify T^2 ~ F(1,10)
curve(df(x,1,10), 0, 20); hist(t_sq, freq=FALSE, add=TRUE)
```

Connections: $\chi^2(d_1)/d_1 \div \chi^2(d_2)/d_2$ • $T \sim t(\nu) \Rightarrow T^2 \sim F(1, \nu)$ • Used in ANOVA and regression tests

Weibull

Continuous

$X \sim \text{Weibull}(\lambda, k)$

Generalises Exponential with a shape parameter k controlling whether the failure rate increases, decreases, or stays constant. The workhorse of reliability engineering and survival analysis.

$$f(x) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x}{\lambda}\right)^k\right), \qquad x \geq 0$$

Mean $\lambda\,\Gamma\!\left(1+\frac{1}{k}\right)$

Variance $\lambda^2\left[\Gamma\!\left(1+\frac{2}{k}\right)-\Gamma\!\left(1+\frac{1}{k}\right)^2\right]$

Parameters:

- $\lambda > 0$: scale — characteristic life
- $k > 0$: shape — controls hazard behaviour

> **Key insight:** Shape k is everything: $k < 1 \to$ decreasing hazard (infant mortality); $k = 1 \to$ constant hazard = Exponential; $k > 1 \to$ increasing hazard (wear-out). Weibull plot: $\log(-\log \hat{S}(t))$ vs. $\log t$ should be linear with slope k—use this to estimate k graphically from survival data.

R:

```
rweibull(1000, shape=k, scale=lambda)
# k=1 is Exponential(1/lambda):
sort(rexp(5,1))[1:3]; sort(rweibull(5,1,1))[1:3]
```

Connections: $k = 1$: Exponential$(1/\lambda)$ • $k = 2$: Rayleigh distribution • Models time-to-failure and lifetimes

Cauchy

Continuous

$X \sim \text{Cauchy}(x_0, \gamma)$

Ratio of two independent $\mathcal{N}(0,1)$ variables—equivalently, $t(1)$. Symmetric and bell-shaped, but with such heavy tails that no moments (mean, variance) exist.

$$f(x) = \frac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}, \qquad x \in \mathbb{R}$$

Mean undefined

Variance undefined

Parameters:

- $x_0 \in \mathbb{R}$: location — median and mode
- $\gamma > 0$: scale — half-width at half-maximum

> **Key insight:** The CLT fails completely: the average of n i.i.d. Cauchy$(0,1)$ variables has the *same* Cauchy$(0,1)$ distribution for all n, because $E[|X|] = \infty$. The bell-shaped appearance is deceptive—extreme values are far more probable than Normal suggests. Cauchy is a stable distribution; it is the archetype for CLT failure.

R:

```
rcauchy(1000, location=x0, scale=gamma)
# Mean does NOT converge:
```

```r
cum_mean <- cumsum(rcauchy(2000))/(1:2000)
plot(cum_mean, type='l')  # never settles
```

Connections: $t(1)$ distribution $= \mathcal{N}(0,1)/\mathcal{N}(0,1)$ • CLT counterexample (stable distribution) • Angular distribution of a spinning turntable